Up or Down? The Optimal Floor to Live On When the Elevator Has to Come and Get You

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Abstract

Which floor of a building minimises the time you spend getting around it? The textbook answer is the middle floor: the median minimises expected distance to a uniformly random destination. But an elevator is not a distance — it is a car that has to come and get you from wherever it was left, most trips start or end at the lobby, other passengers add stops along the way, and for a floor or two the stairs beat waiting. We formalise the question as a facility-location problem with a mechanism, solve it in closed form for one car and no traffic, and validate a discrete-event simulator with real kinematics, collective control, nearest-car dispatch, Poisson background traffic, several cars and a stairs option against it (mean error 2.3%). The middle-floor intuition survives only when nobody goes to the lobby and the car waits where you left it (optimum floor 11 of 20); as soon as a quarter of trips go to the lobby and the car parks there, the optimum drops to floor 2, and at a residential mix ($\alpha = 0.9$) the middle floor costs 163% more time than the best floor and the top floor 381%. Background traffic does not restore the middle: at moderate load the optimum stays low while the curve flattens (regret of the middle 10% at 2 passengers/min), and only at saturation does the optimum drift upward (floor 11 at 4/min), because a car sweeping under load serves the low floors last on its way down. Stairs cut the best floor's trip time from 49 to 38 s by absorbing 10% of trips, and in a lobby-free building they move the closed-form optimum up from floor 5 to 7 — but the up/down asymmetry itself barely matters. A bigger fleet lowers every floor's time yet makes the choice matter *more* in relative terms (regret of the middle 16% $\rightarrow$ 30% from one to four cars), because what remains is ride distance. Across a map of lobby fraction and load, 16 of 24 settings put the optimum on the lowest residential floor; the middle floor is within 5% of optimal only when lobby trips are rare. The practical rule: if you leave the building more than you move within it, live as low as you can, and the elevator's parking policy matters as much as the height.

1 Introduction

Ask which floor of a tall building is the best place to be based — the flat you rent, the office you are assigned — and the reflexive answer is "the middle". The reasoning is the one-median on a line: if destinations are spread uniformly over the floors, the median floor minimises the expected distance to them. It is a clean argument, and it is wrong for most buildings, because it treats the elevator as a distance. An elevator is a mechanism. When you press the button the car is somewhere else — at the lobby where the controller parked it, or wherever it dropped its last passenger — and it has to come to you before it can take you anywhere. Most of your trips do not go to a random floor at all: you leave the building and come back, so the lobby is the destination or origin of the majority of them. Other people are using the same car, and each of them adds a stop to your ride. And when you live on the second floor you do not call the elevator to go down — you take the stairs, which is fast, whereas climbing back up is not.

This paper treats the question as a facility-location problem with a mechanism inside it and answers it quantitatively. We write the expected trip time of a resident on floor h as a function of the lobby-trip fraction α, the elevator's kinematics (rated speed, acceleration, door and transfer times), its idle parking policy (return to the lobby, stay put, park in the middle, or — the heavy-traffic limit — be anywhere), the background passenger rate, the number of cars, and the stairs. For one car and no traffic the expectation has a closed form; for everything else we run a discrete-event simulator with collective control and nearest-car dispatch, validated against the closed form at zero traffic. The objective is the mean trip time, but we also report the 90th percentile, the *regret* of a floor (how much worse than the best), and an *indifference band* (the floors within 5% of the best), because the practical question is as much "does it matter?" as "which floor?".

Four findings organise the paper. (i) The middle-floor answer holds only in the corner where no trip goes to the lobby and the car waits where you left it; with lobby trips and lobby parking the optimum drops to the lowest residential floor and the middle floor is far from indifferent. (ii) Background traffic flattens the curve but does not

move the optimum up until the car is saturated, where an unexpected effect appears: on a loaded car the low floors are the last to be served on the down sweep, so the optimum drifts upward. (iii) Stairs shift the optimum upward by absorbing short trips, but the asymmetry between walking down and climbing up, which we expected to be the driver, contributes little. (iv) More cars make every floor faster and make the choice of floor matter *more*, not less, because what is left of the trip is the ride, which is distance. A map over lobby fraction and load turns these into a rule of thumb.

2 Related work

Elevator traffic analysis. The engineering of lifts has a mature literature on round-trip time, handling capacity and interval — how many cars a building needs and how long people wait — codified in CIBSE Guide D [2] and the treatments of Barney and Al-Sharif [1] and Strakosch [7]. Those analyses take the population's floor distribution as given and size the system; we hold the system fixed and ask the passenger's question. The kinematic model (rated speed, acceleration, door and transfer times) and the collective-control logic we simulate come from that literature.

Facility location on a line. Minimising expected distance to a random point on a line is the 1-median problem, solved by the median of the destination distribution [4]; a fixed "retrieval" cost that depends on where a server is parked turns it into a location problem with a mobile server, close in spirit to the analysis of where to park a single ambulance or repair vehicle [5]. Our closed form is the 1-median with the elevator's parking position added to the distance.

Stairs versus elevator. Studies of stair use in buildings measure walking speeds per flight and the reluctance to climb more than a few floors [3]; the asymmetry between ascending and descending speeds is well documented and is what motivated our stairs model.

Simulation of elevator systems. Discrete-event simulation is the standard tool for evaluating dispatch and group control [6]; our simulator is a minimal member of that family (collective control, nearest-car dispatch, capacity), chosen so that it can be validated exactly against the closed form at zero traffic and so that every effect we report can be traced to a single mechanism.

3 Model

3.1 Building, car and trips

Floors are $1..N$ with 1 the lobby; the resident lives on $h \in \{2..N\}$. A run of n floors takes $nH/v + v/a$ (per-floor cruise at rated speed $v = 1.6\,\mathrm{m/s}$ over $H = 3.3\,\mathrm{m}$, plus one start/stop penalty at $a = 1.0\,\mathrm{m/s^2}$), which coincides with the exact trapezoidal profile whenever a single floor suffices to reach rated speed, as it does here; each stop costs a door cycle of 4 s plus 1.2 s per passenger transferring. The resident alternates *outbound* and *return* trips: outbound goes to the lobby with probability α and otherwise to a uniformly random other floor $f \geq 2$; the return brings them home. Chaining the trips makes the origin/destination pairs exactly symmetric — the car has to fetch you at the lobby as often as at home — and gives a precise meaning to "the car stays where it last stopped".

3.2 Closed form: one car, no traffic

A trip (o, d) costs the car's approach, one dwell, and the ride,

$$T(o, d \mid c) = \mathrm{run}(|c - o|) + \mathrm{stop} + \mathrm{run}(|o - d|), \quad (1)$$

where c is the car's position when called, given by the parking policy: *lobby* ($c = 1$), *middle* ($c = \lceil (N+1)/2 \rceil$), *last* ($c = o$: with no other traffic the car is where the resident got off), or *random* ($c \sim \mathrm{Uniform}\{1..N\}$, the limit in which traffic has left the car anywhere). $E[T(h)]$ sums over the exact distribution of chained (o, d) pairs; with stairs available, each pair costs $\min(\text{elevator}, \text{stairs})$ when the climb is allowed.

3.3 Discrete-event simulator

K cars move floor by floor with the same run and stop costs. A car keeps its direction while there are calls ahead (car calls of passengers on board, hall calls in its direction), reverses when calls remain only behind, and otherwise parks (lobby, middle) or stays (last). A new hall call wakes the nearest idle car; moving cars pick up hall calls they pass in their direction, subject to capacity (10). Background passengers arrive as a Poisson process at λ per minute with the same lobby fraction, travelling up from or down to the lobby, or between two floors ≥ 2. The resident probes the system every two minutes for 4 hours of building time, with 4 replications; replication r uses the same background draw for every home floor (common random numbers), so the curves compare floors rather than traffic realisations. We record the resident's wait, ride, intermediate stops, and the cars' positions at the moment of the call.

3.4 Stairs and objective

Stairs cost $t_\downarrow$ per floor descended and $t_\uparrow$ per floor climbed (defaults 10 and 15 s), with a maximum climb the resident is willing to make (3 floors); the resident walks when the stairs time is below the *expected* elevator time for the trip (they do not know where the car is). For a home floor we report the mean trip time, its 90th percentile, the regret $E[T(h)] - \min_{h'} E[T(h')]$ in percent, and the indifference band of floors within 5% of the best.

4 One car, no traffic: the closed form

Figure 1 is the whole argument in two panels. With no lobby trips ($\alpha = 0$) and a car that stays where it last stopped, the wait is a door cycle and only the ride counts: the optimum is the median floor (11 of 20) and the curve is flat around it. Change either assumption and the optimum moves. If the car parks at the lobby, even with $\alpha = 0$ the approach time grows with h and the optimum slides to floor 7, with an indifference band of floors 3–10. If a quarter of trips go to the lobby, the optimum is floor 2 — the lowest residential floor — and it stays there for every larger α: at $\alpha = 0.5$ the middle floor costs 46% more than the best, at $\alpha = 0.9$ it costs 163% more (14 s versus a top floor at 65 s, 381% worse).

The right panel isolates the mechanism. Parking at the middle floor makes the middle floor optimal at $\alpha = 0.5$ (floor 10), because the approach is now shortest there; the random-position limit gives floor 6 with a wide band (8–13 at $\alpha = 0$), because the approach no longer depends on h and only the lobby trips pull the optimum down; and the stay-put policy gives floor 2 because the resident's own lobby trips leave the car at the lobby half the time. The parking policy, which the resident never chooses, moves the answer as much as the lobby fraction does. The picture scales with height: in a 40-floor building at $\alpha = 0.5$ the optimum with lobby parking is still floor 2 and the middle floor's regret is 67%, while the random-position, no-lobby case puts it at floor 21, the median.

5 Traffic, stairs and fleet

Validation. At zero traffic the simulator reproduces the closed form to 2.3% mean relative error over all floors (Figure 2, left), with the residual coming from the car occasionally being caught mid-return to its parking floor.

Traffic flattens, then inverts. Adding background passengers raises every floor's time and compresses the curve: the regret of the middle floor falls from 83% at zero traffic to 22% at 1/min and 10% at 2/min, while the optimum stays at floor 2. At 4 passengers per minute a single car serving 20 floors is saturated (best mean 160 s, of which most is waiting), and the optimum jumps to floor 11 with a middle-floor regret of only 4%. The middle panel shows why. Under load the car sweeps the building like a bus; a resident on a low floor who wants to go down is the *last* stop of the down sweep, so their wait (148 s on floor 2) exceeds that of a resident on floor 16 (98 s), and only the very top floor, served at the reversal, waits longer still (166 s). Against that, the ride grows with height (26 s to 80 s) and collects up to 5.0 intermediate stops. The two slopes nearly cancel over the middle of the building, which is what "it does not matter" looks like in a saturated system — and the p90-optimal floor (4) remains low, because the tail is made of the sweeps you miss.

Where the car actually is. The right panel measures the stay-put policy under traffic: the car is at the lobby 25% of the time at 0.5/min and 30% at 4/min, with the rest spread almost uniformly. The stay-put policy is, in practice, a mixture of lobby parking and the random-position limit — which is why its optimum behaves like the lobby-parking one at low load and drifts up (floor 16 at 4/min) at high load.

Stairs. With the stairs available (15 s up, 10 s down, at most three floors climbed) the best floor's mean time drops from 49 to 38 s and 10% of trips are walked, almost all of them the one- and two-floor hops of a low-floor resident (Figure 3). The optimum stays at floor 2 at this lobby fraction, and the middle floor's regret *rises* from 22% to 51%, because the stairs help the low floors most. The asymmetry we expected to matter does not: a symmetric 12/12 s stairs walks 10% of trips and a descend-only resident 7%, with the same optimum. Stairs do move the optimum upward in the closed form when lobby trips are absent — from floor 5 to 7 of 15 with lobby parking — but they do so equally with symmetric costs (7) and with descending only (7): the shift comes from the stairs absorbing the short trips that the lobby-parked car served badly, not from the cost of climbing.

Fleet. Adding cars at constant load per car lowers the best floor's time from 65 s (one car) to 52 s (two) and 43 s (four) — and raises the middle floor's regret from 16% to 19% and 30% (Figure 4). More cars remove waiting, and waiting was the part of the trip that did not depend much on h; what remains is the ride, which is distance. The intuition that a well-served building makes the floor irrelevant is backwards: it makes the floor a larger share of what you feel.

6 The map, and does it matter?

Figure 5 puts the pieces on one grid. In 16 of 24 cells — every cell with $\alpha \geq 0.5$ — the optimum is the lowest residential floor, and its indifference band is a single floor. The middle floor is competitive only along the left

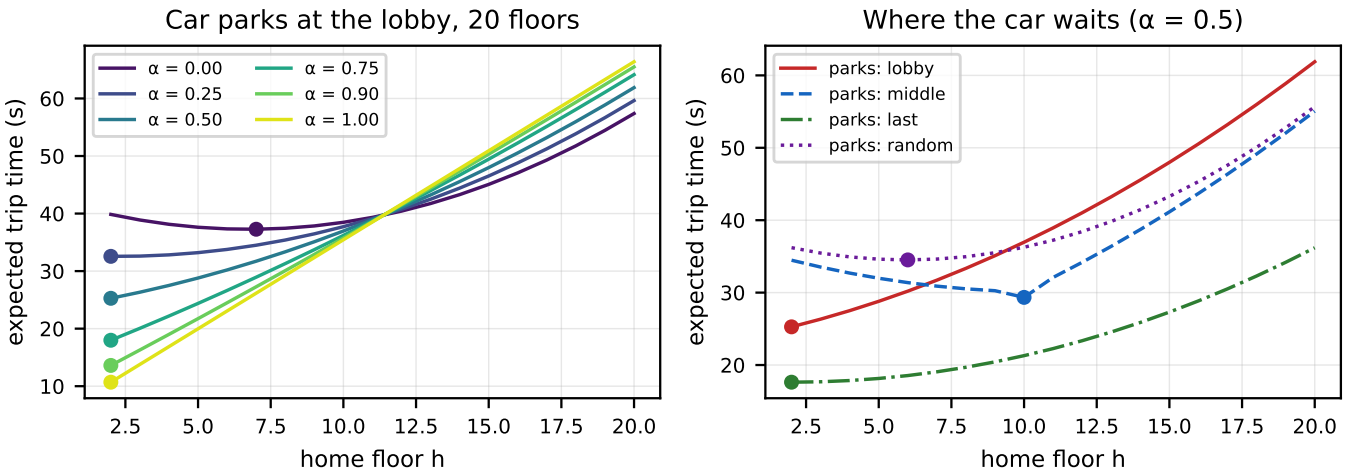


Figure 1: Closed-form expected trip time versus home floor in a 20-floor building. Left: the car parks at the lobby; each curve is a lobby-trip fraction α, dots mark the optimum. Right: $\alpha = 0.5$ under the four parking policies. The middle-floor intuition needs both no lobby trips and a car that waits where you left it.

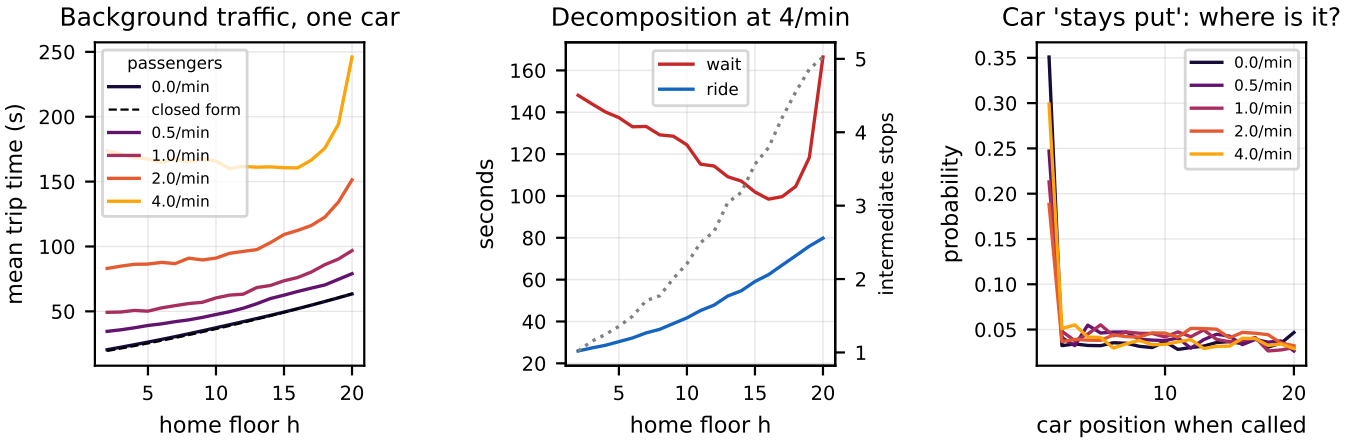


Figure 2: One car, lobby parking, $\alpha = 0.7$, 20 floors. Left: mean trip time versus home floor for background rates from 0 to 4 passengers per minute; the closed form (dashed) matches the simulator at zero traffic. Middle: decomposition at 4/min — the wait falls with height while the ride and the number of intermediate stops rise. Right: where the car is when the resident calls under the stay-put policy: the lobby dominates at every load.

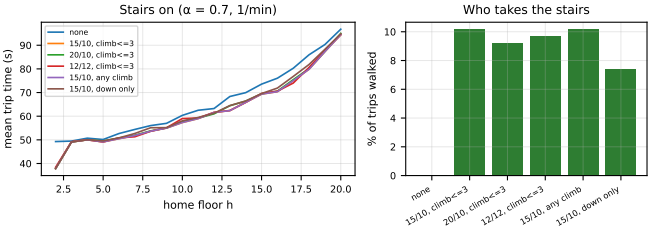


Figure 3: Stairs at $\alpha = 0.7$, 1 passenger/min, one car. Left: mean trip time by home floor for each stairs variant; right: fraction of trips walked.

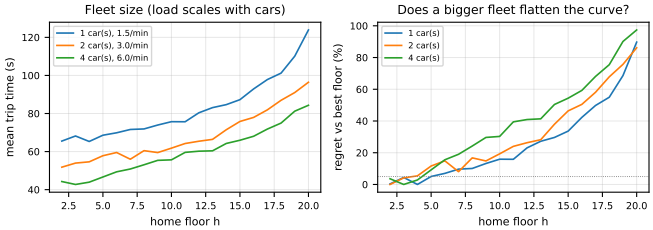


Figure 4: One, two and four cars with a background load that scales with the fleet (1.5 passengers/min per car). Left: mean trip time; right: regret relative to the best floor. A bigger fleet lowers every floor and steepens the curve in relative terms.

edge: at $\alpha = 0$ the optimum is the median (11 at zero traffic, band 7–12) drifting up to 13 under saturation; at $\alpha = 0.25$ it is floor 5 at zero traffic and rises with load to 11, with a band that spans 2–12 at 4/min — the one regime where the honest answer is "it does not matter". Everywhere to the right the middle-floor rule costs tens to hundreds of percent (27% at $\alpha = 0.5$, 96% at $\alpha = 0.9$ with 1/min, where the top floor costs 223%).

Three archetypes make it concrete (Figure 6). A *residential* tower (20 floors, $\alpha = 0.9$, one car, 1/min): optimum floor 2, band 2–2, the middle floor 126% worse and the top floor 288% worse; the p90-optimal floor is the

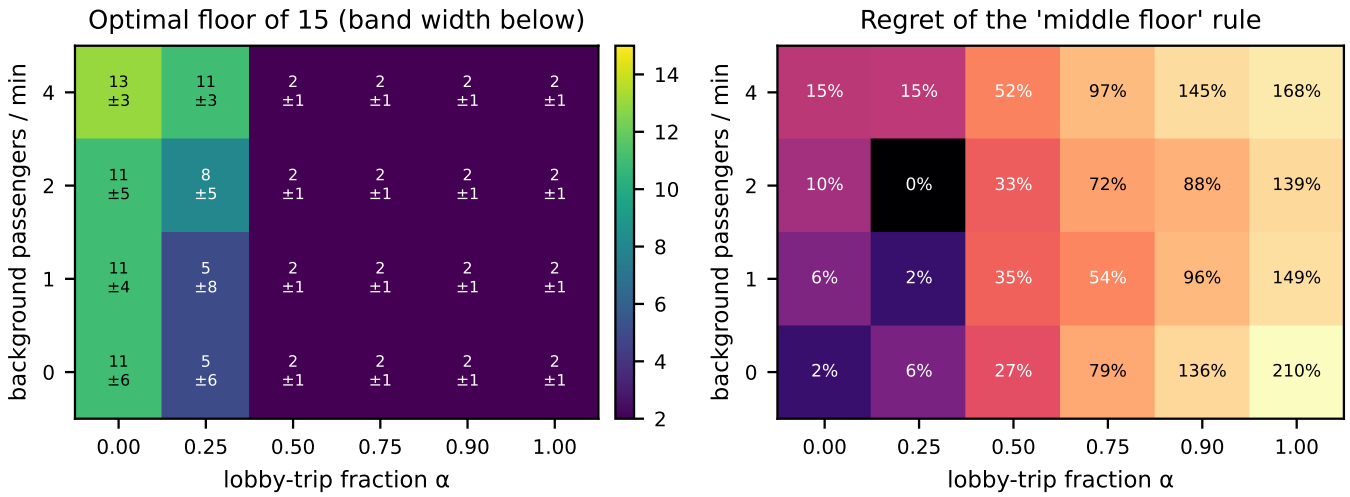


Figure 5: 15-floor building, one car parked at the lobby, stairs on. Left: optimal floor over lobby-trip fraction and background rate, with the width of the 5% indifference band beneath each value. Right: regret of the middle-floor rule in the same cells.

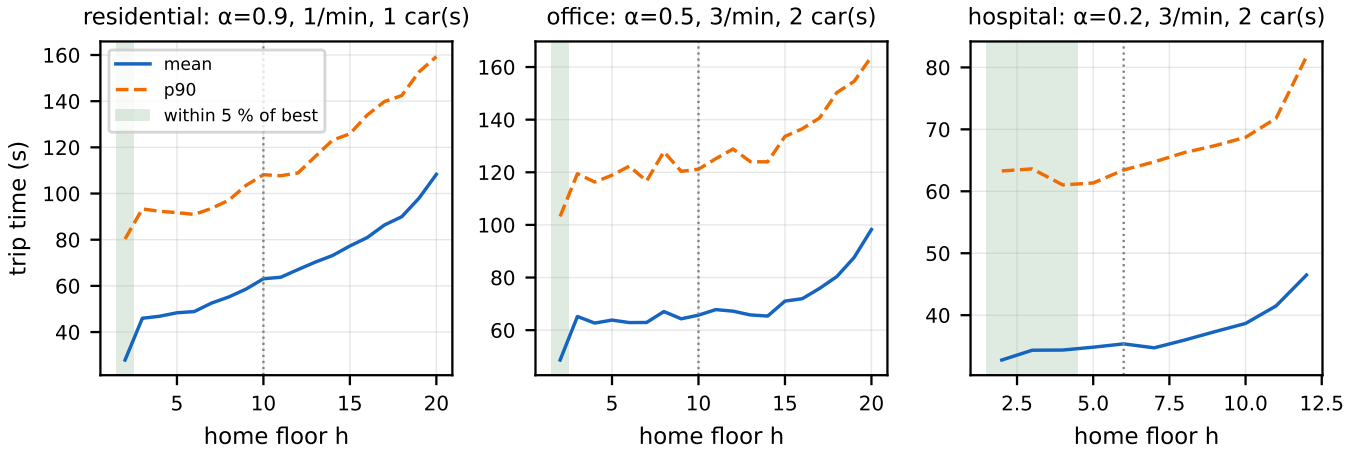


Figure 6: Three archetypes with stairs on: mean and 90th-percentile trip time by home floor, the 5% indifference band (shaded) and the middle floor (dotted).

same. An *office* (20 floors, $\alpha = 0.5$, two cars, 3/min): optimum 2, middle-floor regret 35%, 13% of trips walked. A *hospital*-like building (12 floors, $\alpha = 0.2$, two cars, 3/min, cars parked mid-building): optimum 2 with a band of 2–4, middle-floor regret 8%, p90-optimal floor 4 — the one case where the floors are close to interchangeable and the tail, not the mean, should pick.

The rule of thumb that survives all of this: count how often you leave the building against how often you move within it. If leaving dominates, the lowest floor you can get wins by a wide margin, the elevator's parking policy is the next most important thing, and the stairs make the low floor better still. If moving within dominates and the cars are busy, live anywhere from the middle up and stop worrying.

7 Limitations and conclusion

Limitations. The building is a stylised one: uniform floor population, a single lobby, no express zones or sky lobbies, no time-of-day peaks, and a resident whose destinations are uniform over the other floors. The simulator implements collective control with nearest-car dispatch, not a modern destination-dispatch group controller, and its parking policies are the three textbook ones; real controllers learn where to park, which would move our "lobby" and "random" limits toward each other. The kinematics are mid-rise values (1.6 m/s), so the floor-by-floor run model coincides with the trapezoid; high-speed cars would need the exact profile, which the code provides. The stairs rule is deterministic on expectations and ignores effort, carrying things, and the building's actual stair placement. The resident's probing trips are frequent enough to perturb a loaded single car slightly. All of these are configuration knobs, and the accompanying full-scale configuration widens every grid.

Conclusion. The middle floor is the right answer to a question nobody asks — where to be if every trip is to a random floor and the elevator is waiting outside your

door. Once the elevator has to come and get you, the answer depends on where it waits; once most trips go to the lobby, the answer is the lowest floor you can get, by a margin that grows with the building and with the number of cars. Traffic blurs the curve but does not move the optimum up until the car is saturated, and there the effect that lifts it is a queueing one (the low floors are served last on the down sweep) rather than a geometric one. Stairs make the low floors better still and shift a lobby-free building's optimum upward by absorbing short trips, but the cost of climbing is not what does it. And the sense in which "it does not matter" is precise: the indifference band is one floor wide wherever lobby trips are common, and spans most of the building only where they are rare and the car is busy. Up or down? Down, unless you never leave.

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